

MA.6.NSO.2.2

Benchmark

MA.6.NSO.2.2 **Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions, including mixed numbers, with procedural fluency.**

Benchmark Clarifications:

Clarification 1: Instruction focuses on making connections between visual models, and the relationship between multiplication and division, reciprocals and algorithms.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.3.3
- MA.6.AR.1.1, MA.6.AR.1.3
- MA.6.GR.2
- MA.6.DP.1.2, MA.6.DP.1.3

Terms from the K-12 Glossary

- Area Model
- Commutative Property
- Dividend
- Divisor
- Expression

Vertical Alignment

Previous Benchmarks

- MA.5.FR.2.2, MA.5.FR.2.4

Next Benchmarks

- MA.7.NSO.2.2

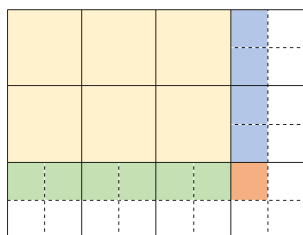
Purpose and Instructional Strategies

In grade 5, students multiplied fractions by fractions with procedural reliability and explored how to divide a unit fraction by a whole number and a whole number by a unit fraction. In grade 6, students become procedurally fluent with multiplication and division of positive fractions. The expectation is to utilize skills from the procedural reliability stage to become fluent with an efficient and accurate procedure, including a standard algorithm. In grade 7, students will become fluent in all operations with positive and negative rational numbers.

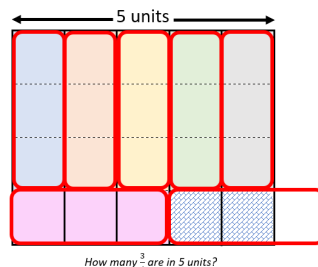
- Instruction includes making connections to the distributive property when multiplying fractions.
 - For example, when multiplying $1\frac{1}{2}$ by $\frac{3}{4}$, it can be written as $\left(1 + \frac{1}{2}\right)\frac{3}{4}$ to determine $\frac{9}{8}$ as the product.
- Instruction includes making connections to inverse operations when multiplying or dividing fractions.
 - For example, when determining $\frac{3}{4} \div \frac{5}{8}$, students can ask the question, what number multiplied by $\frac{5}{8}$ gives the value $\frac{3}{4}$?

- Instruction focuses on appropriate academic vocabulary, such as reciprocal. Avoid focusing on tricks such as “keep-change-flip.” Using academic language and procedures allows students to connect to future mathematics (*MTR.5.1*).
 - For example, $\frac{3}{4} \div \frac{5}{8}$ can be read as “How many five-eighths are in three-fourths?”
- Instruction includes using concrete and pictorial models, writing a numerical sentence that relates to the model and discovering the pattern or rules for multiplying and dividing fractions by fractions (*MTR.2.1*, *MTR.3.1*, *MTR.5.1*).
 - Area Model

$$2\frac{1}{2} \times 3\frac{1}{2} = 8\frac{3}{4}$$

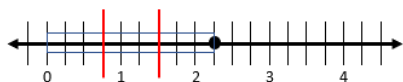


$$5 \div \frac{3}{4} = 6\frac{2}{3}$$

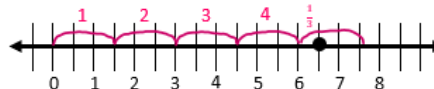


- Linear Model

$$2\frac{1}{4} \times \frac{2}{3} = 1\frac{1}{2}$$

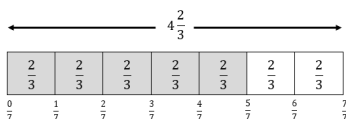


$$6\frac{1}{2} \div 1\frac{1}{2} = 4\frac{1}{3}$$

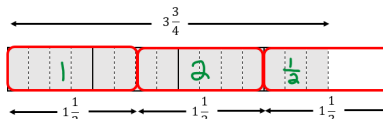


- Bar Model

$$4\frac{2}{3} \times \frac{5}{7} = 3\frac{1}{3}$$



$$3\frac{3}{4} \div 1\frac{1}{2} = 2\frac{1}{2}$$



- Instruction includes providing opportunities for students to analyze their own and others' calculation methods and discuss multiple strategies or ways of understanding with others (*MTR.4.1*).
- Students should develop fluency with and without the use of a calculator when performing operations with positive fractions.

Common Misconceptions or Errors

- Students may forget that common denominators are not necessary for multiplying or dividing fractions.
- Students may incorrectly assume that multiplication results in a product that is larger than the two factors. Instruction continues with students assessing the reasonableness of their answers by determining if the product will be greater or less than the factors within the given context.
- Students may incorrectly assume that division results in a quotient that is smaller than the dividend. Instruction continues with students assessing the reasonableness of their

answers by determining if the quotient will be greater or less than the dividend within the given context.

Strategies to Support Tiered Instruction

- Teacher encourages students with a firm understanding of multiplying and dividing decimals to convert fractional values to their equivalent decimal forms and then perform the desired operation and convert the solution back to fractional form.
- Instruction includes the use of fraction tiles, fraction towers, or similar manipulatives to make connections between physical representations and algebraic methods.
- Instruction includes the co-creation of a graphic organizer utilizing the mnemonic device Same, Inverse Operation, Reciprocal (S.I.R.) for dividing fractions, which encourages the use of correct mathematical terminology, and including examples of applying the mnemonic device when dividing fractions, whole numbers, and mixed numbers.
- Teacher provides students with flashcards to practice and reinforce academic vocabulary.
- Instead of multiplying by the reciprocal to divide fractions, an alternative method could include rewriting the fractions with a common denominator, dividing the numerators and then dividing the denominators.
 - For example, $\frac{5}{6} \div \frac{3}{2}$ is equivalent to $\frac{5}{6} \div \frac{9}{6}$ which is equivalent to $\frac{5/9}{1}$ which is equivalent to $\frac{5}{9}$.
- Instruction provides opportunities to assess the reasonableness of answers by determining if the product will be greater or less than the factors within the given context.
- Instruction provides opportunities to assess the reasonableness of answers by determining if the quotient will be greater or less than the dividend within the given context.

Instructional Tasks

Instructional Task 1 (MTR.2.1, MTR.4.1)

Jasmine wants to build a $2\frac{5}{6}$ meters long garden path paved with square stones that measure $\frac{1}{4}$ meter on each side. There will be no space between the stones.

Part A. Create a model that could be used to answer the following question: How many stones are needed for the path?

Part B. How many stones are needed for the path?

Instructional Task 2 (MTR.3.1, MTR.6.1)

A container at a juicing plant holds $6\frac{2}{3}$ tons of oranges. The plant can juice $1\frac{1}{2}$ tons of oranges per day. At this rate, how long will it take to empty the container?

Instructional Task 3 (MTR.2.1)

Explain using visual models why $\frac{4}{5} \times \frac{2}{3} = \frac{8}{15}$.

Instructional Items

Instructional Item 1

What is the value of the expression $\frac{3}{5} \div \frac{5}{8}$?

Instructional Item 2

What is the value of the expression $8\frac{1}{10} \div \frac{5}{8}$?

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*